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Research and Implementation of Road Damage Detection Algorithm Based on Object Detection Network

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***Abstract*—Various types of road damage occur frequently, which can affect the smooth running of vehicles. The detection of road surface damage is of great significance for road surface maintenance and smooth traffic flow. First, this paper makes descriptive statistics on RDD2020 dataset, and deals with the mislabeled categories in the dataset, through which 14,569 samples are obtained. A single-stage object detection network YOLOv5 is then constructed to detect road damage on the RDD2020 dataset. The experiment results show that the proposed network is effective in road damage detection of RDD2020 dataset. Faced with high-cost detection methods, a convenient and efficient road damage detection network is urgently needed. In this paper, a road damage detection system is deployed, which can detect the location of road damage and identify the types of road damage in real-time under the camera shooting.**

Keywords-RDD2020 Dataset; Road Damage Detection System; YOLOv5 Network

I. INTRODUCTION

To establish an efficient and accurate road damage detection network, it is necessary to consider utilizing fast detection methods to determine the latest road health. In this paper, the target detection method in artificial intelligence technology is used to solve this problem, so as to treat different damages and repair them. This has a positive effect on reducing road pressure and improving people's lives. Through the combination of image processing and deep learning direction, the damage on the road is detected by the method of target detection, thus obtaining a reliable road damage detection network, which can detect the damage of unknown pavement situations. The network can be deployed to detect real-time video, thus further improving the use of the network for road damage and ease of detection.

By generally introducing the RDD2020 dataset, this paper found that the dataset contains a large amount of suitable high-quality data, and it meets the requirements of classification and recognition for different road damage situations. Moreover, there are numerous types of road damage in the images and the data sources are extensive, such that the problem can be turned into a multi-class object detection problem and the appropriate network needed for training can be found. Overall, the contributions of this paper are as follows:

- The YOLOv5 road damage model we generated can cope with most road damage situations, and can be directly invoked and deployed through different methods.
- Our user visualization interface can be modified according to requirements to achieve the desired effect.
- Our executable files are capable of image, video, and real-time video detection, and can be directly used for actual road detection, or deployed through vehicles, and road damage detection through external cameras.

II. RELATED WORK

Arya et al. [1] introduced detailed information on the RDD2020 dataset, which contains a great number of road images from different national institutions, including more than 31,000 examples of road damage. This dataset captures various kinds of road damage, respectively: Longitudinal Cracks, Transverse Cracks, Alligator Cracks, and Potholes. The purpose is to develop a new automatic detection method for road damage classification through deep learning.

Everingham et al. [2] pointed out the limitations and weaknesses of the current generation of algorithms, and introduced a guidance method to determine whether the performances of the two algorithms are significantly different. In addition, the normalized average accuracy was introduced, so that the performance can be compared between classes with different proportions of positive and negative instances. Then, a clustering method for visualization and spanning multiple

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algorithms was introduced, so that illegible images could be identified. Furthermore, a joint classifier was used in the submitted algorithms to measure their complementarity and combination performance. Finally, the challenges that can be improved in the future were proposed.

Arya et al. [3] evaluated the availability of a low-cost and easy-to-use smartphone-based automatic road condition monitoring method developed in Japan. They presented a general model that can detect and classify road damage in many countries. On the other hand, Arya et al. [4] pointed out that the network with better performance uses integrated learning based on YOLO network [5], and reviewed the aspects that are effective for the current challenges in the field of target detection and the aspects that can be improved in the future challenges.

Maeda et al. [6] researched road surface damage detection by using image processing technology. They used the most advanced object detection method in convolutional neural networks, and employed their own collected datasets to train road damage detection networks, which were also deployed in the server and mobile terminals respectively. The experimental results demonstrated that road damage can be accurately divided into eight types by using their target detection method. Meanwhile, Maeda et al. [7] utilized a generation confrontation network to generate a false image that cannot be distinguished from the real image, thus enhancing the data. In addition, the progressive generation confrontation network and Poisson hybrid method are combined to artificially generate road damage images, which can be adopted as new training data to improve the accuracy of road damage detection.

Hong [8] analyzed and compared the road damage detection systems, introduced the present situation of road damage detection systems, and proposed a BLT road damage detection system.

Huang et al. [9] described a series of contents such as pavement subgrade engineering, including the development of pavement subgrade engineering, maintenance and management of design subgrade pavement, etc.

Rezatofighi et al. [10] presented the evaluation criteria in target detection, and introduced a generalized version of IoU into the target detection framework as a new loss function and evaluation index. It was used to solve the weakness of IoU in the case of non-overlapping borders, called GIoU, and showed the performance after the improvement as well as its performance on VOC and COCO datasets.

However, the traditional road condition evaluation method has problems such as consuming a lot of labor, high time consumption, high cost, strong subjectivity, and prone to inspection errors. And with the limited number of specialists who can assess such road damage, undetected damaged roads cannot be maintained in a timely manner, making the problem of pavement damage even worse. In addition, many local governments are unable to conduct adequate road health inspections on time and as needed due to lack of financial resources. And some municipalities use high-performance sensors to automatically detect road damage. However, due to the large area of roads that need to be inspected, the many road branches, and the high cost of these sensors, it is not feasible to use them on all roads. Therefore, there is an even more urgent need for a system that can easily assess road conditions and identify road surface damage at low cost and in less time.

This paper solves the problem through the combination of traditional road survey and artificial intelligence technology. Thereby facing different damages, symptomatic repair. Through the combination of traditional images and annotations and deep learning knowledge, the method of target detection is used to train the damage on the road to obtain a reliable prediction model, and then the final weight is obtained by adjusting the model file, which can detect unknown road surface damage, and then package the operation through corresponding deployment, so that it can become a convenient application program, so that it can detect image files, video files or real-time video detection, to further improve the model for road The purpose of damage and the ease of detection.

III. RDD2020 Dataset

A. Introduction to Dataset

The RDD2020 dataset contains 26,336 road images collected from India, Japan and the Czech Republic, with more than 31,000 examples of road damage. There are four types of road damage in the dataset namely Longitudinal Cracks, Transverse Cracks, Alligator Cracks and Potholes.

B. Statistical Description of Dataset

Due to the diversity of labelling institutions and the uncertainty of manual labelling, there are some problems in data labelling, such as blank labelling, mislabeling, and extra labelling. The statistical results are presented in Fig. 1. It can be seen from Fig. 1 that the data given by Japanese institutions is good, and the qualified rate of both data and labelling information is high, while the data given by Czech institutions is inferior, and many unqualified data are found. The qualified data of Indian institutions accounts for about half of the total.

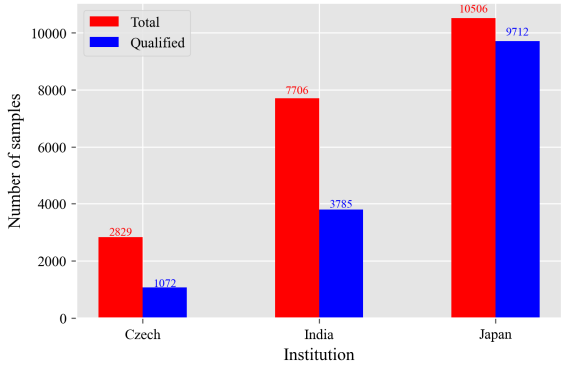


Figure 1. Data filtering situation.

According to the specific situation of image categories, summarize the distribution of labelling information in each organization, as shown in Fig. 2, where LC is the abbreviation of Longitudinal Cracks, TC is the abbreviation of Transverse Cracks, AC is the abbreviation of Alligator Cracks and P is the abbreviation of Potholes. From the histogram of dataset categories in Fig. 2, it is easily found that the total amount of category labels are considerably higher than the total sample size

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of dataset images, which indicates that the images in the dataset contain more than one category, that is, the road images contain one or more road damages. Therefore, this dataset is not a single-classification target detection problem, but a multi-classification target detection problem. Moreover, the original dataset contains ten different categories, but only four specific categories and names are given in the data introduction. This situation is attributed to the fact that an additional six different categories, names, and wrong location labels are present in the label information.

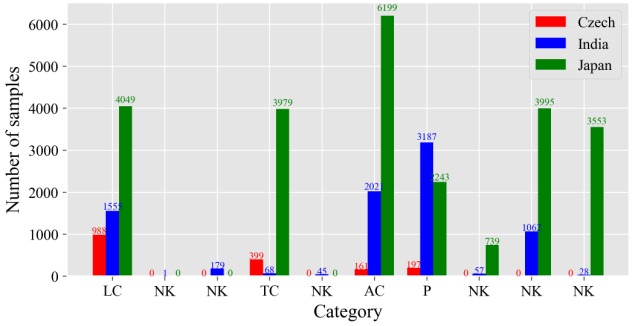


Figure 2. Category of original dataset.

As each image contains more than one kind of classification, it is speculated that there are three specific reasons for this situation: (1) wrong classification, that is, judgment error occurs in the process of category determination; (2) superclass classification means that the category that does not belong to the dataset is added; (3) Blank category refers to only the basic information of the image, but without specific labelling information. In addition, the number of other redundant classifications is significantly smaller than the number of correct four classifications. In view of the above situation, this paper decided to remove the anomalous label information of the mistakenly classified and super-classified images while only retaining the accurately classified label information whereas directly deleting the blank images.

Reviewing the categories of the processed datasets, the data distribution bar chart is shown in Fig. 3. The processed data has a total of 14,569 images, each of which contains a varying number of tags of four categories: Longitudinal Cracks, Transverse Cracks, Alligator Cracks and Potholes. The size of each image is 600×600.

IV. Methodology

A. YOLOv5 Network

In this paper, YOLOv5 is used as a network for road damage detection, and the network structure is shown in Fig. 4. The loss function of YOLOv5 network consists of three parts, namely Classification Loss, Localization Loss, and Confidence Loss. The total loss is the weighted sum of the three losses, and the importance of the three losses can be adjusted by changing the

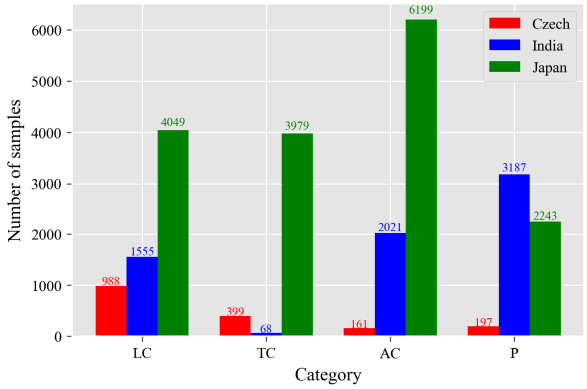


Figure 3. The category of dataset after processing.

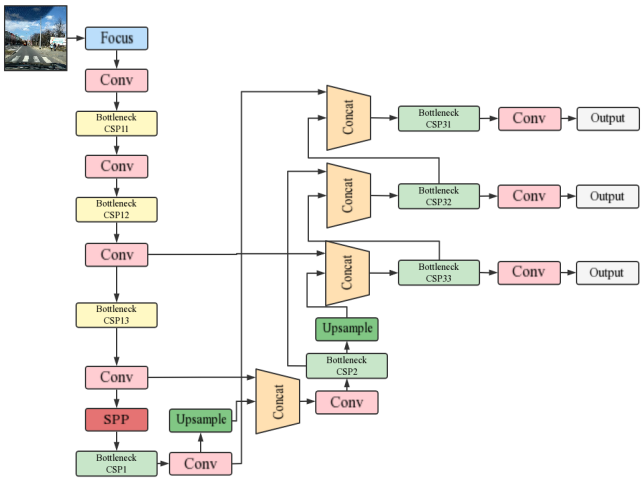


Figure 4. YOLOv5 network structure.

weights, which has a huge impact on the accuracy of target detection.

Classification Loss: The classification loss of YOLOv5 network is calculated by using binary cross entropy function, and the formula shows as follows:

$$y_i = \text{Sigmoid}(x_i) = \frac{1}{1+e^{-x_i}} \tag{1}$$

$$L_{\text{class}} = -\sum_{n=1}^{N} y_i^* \log(y_i) + (1-y_i^*)\log(1-y_i) \tag{2}$$

where N is the total number of categories, x_i is the predicted value of the current category, y_i is the probability of the current category after activating the function, y_i^* is the real value of the current category, and L_{class} is the classification loss.

Localization Loss: YOLOv5 network adopts GIoU function to calculate localization loss, and the formula is as follows:

$$S_3 = (\max(x_2^p, x_2^g) - \min(x_1^p, x_1^g)) \cdot (\max(y_2^p, y_2^g) - \min(y_{p_1}, y_1^g)) \tag{3}$$

$$\text{GIoU} = \text{IoU} - \frac{S_3 - S_2}{S_3} \tag{4}$$

$$L_{\text{GIoU}} = 1 - \text{GIoU} \tag{5}$$

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where $(x_1^p, y_1^p, x_2^p, y_2^p)$ is the predicted frame coordinate and $(x_1^g, y_1^g, x_2^g, y_2^g)$ is the real frame coordinate. S_3 is the area of the smallest rectangular frame surrounding the prediction frame and the real frame, and S_2 is the area of the union area of the prediction frame and the real frame. IoU is the intersection ratio of real frame and prediction frame to the union of real frame and prediction frame.

Confidence Loss: YOLOv5 network applies FocalLoss strategy to solve the problem of serious imbalance between positive and negative samples in target detection so as to calculate the confidence loss. The formula is as follows:

$$\mathrm{FL}(p_t) = -(1-p_t)^{\gamma}\log(p_t) \tag{6}$$

B. Evaluating Indicator

In this paper, Precision, Recall, and mAP are used to evaluate YOLOv5 network. Precision and Recall have the following formula:

$$\text{Precision} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \tag{7}$$

$$\text{Recall} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \tag{8}$$

where TP is the amount of positive samples correctly identified as positive samples, TN is the amount of negative samples correctly identified as negative samples, FP is the amount of negative samples wrongly identified as positive samples, and FN is the amount of positive samples wrongly identified as negative samples.

MAP@0.5 means that when IoU is 0.5, all classes are calculated separately to get their AP, and then the average calculation is made. MAP@0.5 is also the area under the PR curve. MAP@.5:.95 means that when IoU starts from 0.5 and steps from 0.05 to 0.95 each step the obtained ten mAPs are averaged.

V. Experiment

A. Experimental Parameters and Environment Settings

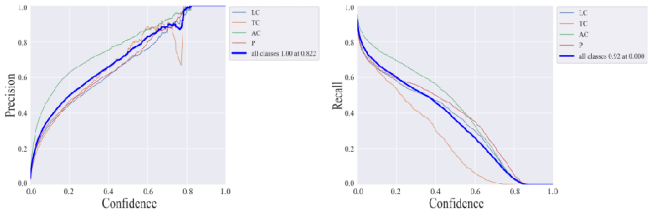
The hardware configuration used in this paper is a 3.70 GHz 10-core Core i9-10900X CPU with two NVIDIA GeForce RTX 2080 Ti graphics cards and 64G running memory. The network training parameters are shown in Table 1.

TABLE 1. Training Parameter Settings

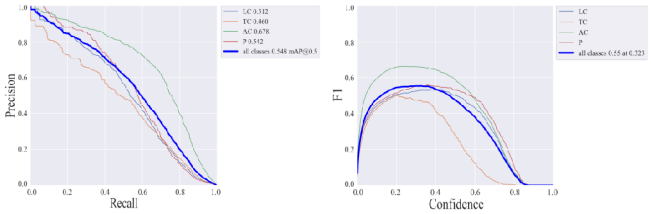
Parameter	Settings
Input Size	640×640
Batch Size	16
Training Epoch	100
Initial Learning Rate	0.01
Final Learning Rate	0.2
Momentum Factor	0.937
Weight Attenuation Coefficient	5e-4

B. Model Training and Evaluation

Using RDD2020 dataset to conduct training on YOLOv5 network, the trained network is evaluated by drawing Precision curve, Recall curve, Precision and Recall (PR) curve and Score curve on RDD2020 test set, as shown in Fig. 5. It can be seen that when Recall drops to 0, the average Precision of the network is 1, and the Precision of the individual categories is also 1. Precision also showed a downward trend as Recall increased. However, when Recall is 1, the Precision of the model is 1. By looking at the overall performance of the PR curve and looking at the area in the lower left of the PR curve, it is found that the area on Alligator Cracks is the largest, and the area on Transverse Cracks is the smallest, with a total area of more than 50%.



(a) Precision Curve (b) Recall Curve



(c) PR Curve (d) F_1 Curve

Figure 5. Training process change curve.

C. Model Prediction

With the trained model, the test set of RDD2020 is examined and the detailed test results parameters are shown in Table 2. It can be seen that the average value of mAP@.5 is 0.548, and the mAP@.5 of different categories is between 0.46 and 0.68, indicating that there is still more space for optimization of the network. The lowest mAP@.5 of Transverse Cracks is only 0.460. This is due to the apparent sample imbalance of the dataset and the small sample size of Transverse Cracks, which results in an unsatisfactory predictive effect for this category.

TABLE 2. Test Result Parameters

Category	Number of samples	Precision	Recall	mAP @.5	mAP @.5:.95
All	4916	0.604	0.517	0.548	0.233
Longitudinal Cracks	1305	0.552	0.510	0.512	0.211
Transverse Cracks	914	0.586	0.399	0.460	0.173
Alligator Cracks	1697	0.713	0.612	0.678	0.325
Potholes	1000	0.565	0.546	0.542	0.223

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The confusion matrix of YOLOv5 after training is shown in Fig. 6. It can be seen that most of the damage categories are incorrectly predicted as Longitudinal Cracks, and some of the categories are predicted as background. In addition, some Potholes were incorrectly predicted as Alligator Cracks. However, there is no large-scale misjudgment, so the network effect is good.

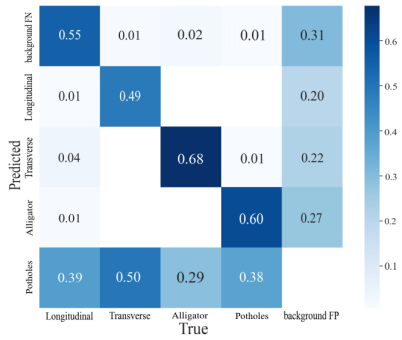


Figure 6. Confusion matrix.

D. Model Deployment

Next, use GUI to realize the deployment of the network. Therefore, it is more convenient and faster to apply the network to implement practical functions and solve practical production problems. GUI, also known as graphical user interface, is a visual approach to display an operational interface for user convenience. Compared with the command line operation, the graphical interface is more acceptable visually.

This system provides three detection modes: picture detection, video detection, and real-time monitoring of camera calls. First of all, design a GUI interface, and the user graphical interface is shown in Fig. 7. The network successfully detected the road damage in the picture to be detected, and marked the damage category and classification confidence on the picture. In the image detection results, it can be seen that the network is able to recognize the damage on the actual road. In video detection, the network can smoothly detect 1080p 60 frames of video, as well as real-time detection of road damage during video playback. Moreover, in the camera-shot mode, the network can smoothly perform real-time video inference and synchronously identify the category of damage on the road and the location of the damage.

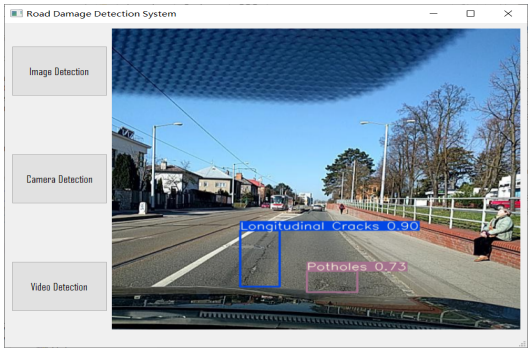


Figure 7. User graphical interface.

VI. Conclusion

Road damage detection is an indispensable task, and only by continuously combining with the latest technology can an effective solution be obtained. In this paper, the method of combining image processing with target detection is used to solve the problem of road damage detection. The trained network has achieved good results, and through the local test and the detection of actual roads found that the generalization of the network is excellent, which can be used as a method to solve such problems.

In the future, there may be more excellent solutions to such problems. Only by constantly assimilating new knowledge, making it applicable to existing problem scenarios, and continuously optimizing previous methods, can the solution to the problem be more superior and accurate, serving people better, and thus promoting social and economic development.

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